








Statistics of Financial Markets

Chapter 8: Volatility estimators



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Acronyms used in this material (1/2)

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); Abnormal Return
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARCH-LM**: ARCH Lagrange Multiplier (test)
- **BET**: Bucharest Exchange Trading (index)
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **CBOE**: Chicago Board Options Exchange
- **DM**: Diebold–Mariano (test)
- **EODHD**: EOD Historical Data (market-data provider)
- **ES**: Expected Shortfall
- **EWMA**: Exponentially Weighted Moving Average
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity

Acronyms used in this material (2/2)

- **GK**: Garman–Klass (volatility estimator)
- **HAR**: Heterogeneous AutoRegressive (model)
- **IGARCH**: Integrated GARCH
- **LM**: Lagrange Multiplier
- **MSE**: Mean Squared Error
- **MZ**: Mincer–Zarnowitz (regression)
- **OHLC**: Open, High, Low, Close
- **OMV**: Österreichische Mineralölverwaltung — Austrian Mineral Oil Administration (OMV group)
- **QLIKE**: Quasi-LIKElihood (loss function)
- **RV**: Realised Variance
- **SE**: Standard Error
- **SFE**: Statistics of Financial Markets, Exercises (Quantlet collection of the textbook)
- **VIX**: CBOE Volatility IndeX
- **YZ**: Yang–Zhang (volatility estimator)

Today's Question and Route

- **Question**: how large is the risk of an asset today, if we can never observe it directly?
  - Chapter 7: returns are almost unpredictable, but their size is not; here we measure that size
  - Chapter 9 turns the measurements into a model (ARCH and GARCH)
- **Route** of the chapter
  - what volatility is and why it is latent; historical volatility and the choice of the window
  - EWMA (RiskMetrics); range-based estimators from open, high, low and close prices
  - realised volatility; volatility clustering and its tests
  - long-run behaviour, the VIX, the leverage effect; how to compare volatility forecasts

Learning Outcomes

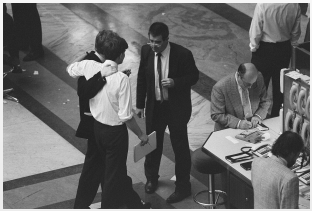
- Define volatility, annualise it with the actual trading frequency and explain why it is a latent quantity
- Compute historical and EWMA volatility and choose a window or a decay factor
- Compute the Parkinson, Garman–Klass, Rogers–Satchell and Yang–Zhang estimators and state their assumptions
- Detect volatility clustering with the ACF of squared returns, the Ljung–Box test and the ARCH-LM test
- Compare implied and realised volatility, and evaluate volatility forecasts with MSE and QLIKE

Reading and Tools

- Textbook: Franke, Härdle & Hafner (2019), Ch. 13 (introduction: volatility and conditional heteroskedasticity)
  - exercises with solutions: Borak, Härdle & López-Cabrera (2013), Ch. 13
- Volatility and its stylised facts: Tsay (2010), Ch. 3.1; a review of volatility forecasting: Poon & Granger (2003)
- Python Quantlets of this chapter: Quantlets/Ch_08
  - ported from the SFE Quantlets SFEkurgarch (kurtosis of a GARCH process) and SFEvolnonparest (nonparametric volatility)
- Lecture notebook: open in Google Colab
- Video course: Applied Time Series Analysis with Python

Volatility Comes in Bursts

- "Large changes tend to be followed by large changes, of either sign, and small changes tend to be followed by small changes" Mandelbrot (1963)
  - after the crash of 19 October 1987, markets stayed turbulent for weeks
- Volatility enters almost every number in finance
  - risk measures: VaR and ES (Chapter 10), the Sharpe ratio (Chapter 1)
  - option prices, margins of exchanges, the size of positions
- This chapter: how to **measure** it from prices, and how to tell whether it clusters



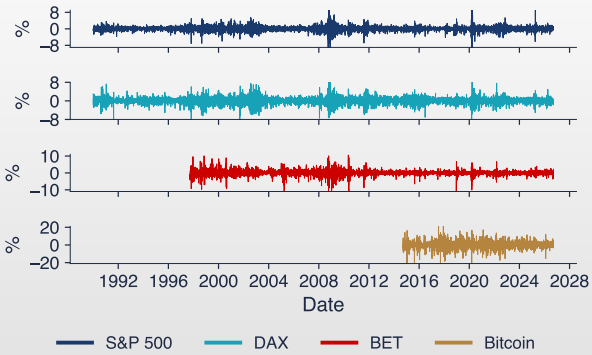
Amsterdam Stock Exchange, 21 October 1987, two days after the crash

Photo: Bart Molendijk / Anefo (1987); CC0; Wikimedia Commons

Volatility: Definition and Units

- Daily log return in %: $r_t = 100\,(\ln P_t - \ln P_{t-1})$ (Chapter 1)
  - **volatility** = the standard deviation of returns, $\sigma = \sqrt{\mathrm{Var}(r_t)}$
  - it has the units of the return (% per day); the variance σ^2 has units of $\%^2$
- **Annualisation**: with A observations per year and uncorrelated returns, $\sigma_{\text{year}} = \sigma_{\text{day}}\sqrt{A}$
  - A = the actual frequency: about 252 for the S&P 500, 250 for the BET, 365 for Bitcoin (trades every day)
  - the rule $\sqrt{A}$ is the square-root-of-time rule of Chapter 7: it needs uncorrelated returns
- Volatility measures dispersion in both directions: a large gain counts as much as a large loss
  - other measures of uncertainty exist, for example the information entropy of returns Pele, Lazar & Dufour (2017)

Four Markets, One Pattern



- Daily log returns in %: S&P 500 and DAX since 1990, BET since 1997, Bitcoin since 2014; last day 18 September 2026
- Quiet years alternate with turbulent months in all four series

SFM_ch8_returns_volatility

Interpretation: Volatility in Four Markets

	Annual volatility	Largest daily move	Date	Excess kurtosis
S&P 500	18.0%	−12.8%	16 March 2020	10.9
DAX	21.8%	−13.1%	12 March 2020	6.0
BET	23.4%	−13.1%	7 January 2009	9.9
Bitcoin	66.6%	−46.5%	12 March 2020	12.0

- The largest moves are 8 to 13 standard deviations: practically impossible under the Normal distribution (Chapter 2)
- Excess kurtosis far above 0 (the Normal value): heavy tails; Section 6 shows that clustering is one cause

SFM_ch8_returns_volatility

Volatility Is Latent

- Model: the return is a constant mean plus a shock whose size changes every day: $r_t = \mu + \sigma_t z_t$
  - μ: the mean return; z_t: i.i.d. shocks with mean 0 and variance 1
  - σ_t is the **conditional volatility**: $\sigma_t^2 = \mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$, with $\mathcal{F}_{t-1}$ the information at the close of day $t-1$
  - the **unconditional volatility** $\sigma = \sqrt{\mathrm{Var}(r_t)}$ is its long-run average level
- **Latent**: we never observe σ_t, only one return per day
  - r_t^2 is an unbiased but very noisy measure of σ_t^2 (with $\mu = 0$): $E[r_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2$
  - a quiet day can happen in a turbulent period, and the other way round
- Three ways to get more information about σ_t
  - average many days: historical volatility, EWMA
  - use the whole path of the day: the high and low prices
  - use intraday prices: realised volatility

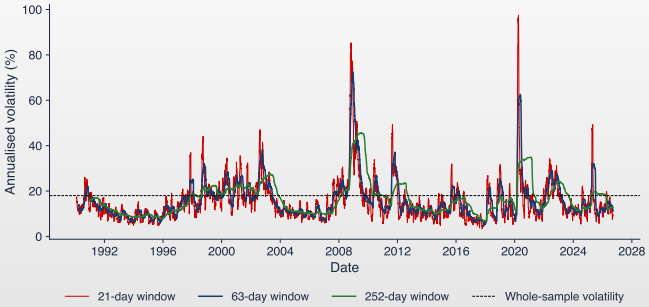
Worked Example: Annualising Volatility

- S&P 500 since 1990: daily standard deviation $\hat{\sigma} = 1.1337\%$, $A = 251.7$ observations per year (the actual frequency)
  - $\hat{\sigma}_{\text{year}} = 1.1337 \times \sqrt{251.7} = 1.1337 \times 15.866 = 18.0\%$
- Bitcoin since 2014: $\hat{\sigma} = 3.4852\%$ per day, 7 days a week
  - correct: $3.4852 \times \sqrt{365.3} = 3.4852 \times 19.114 = 66.6\%$
  - wrong, with the equity convention: $3.4852 \times \sqrt{252} = 3.4852 \times 15.875 = 55.3\%$
- Comparing assets: annualise each with its own calendar, then compare

Historical (Rolling) Volatility

- **Historical volatility** on a window of the last n days:
  - $\hat{\sigma}_t = \sqrt{\frac{1}{n-1}\sum_{j=0}^{n-1}(r_{t-j} - \bar{r}_t)^2}$, with $\bar{r}_t$ the mean of the same n returns
  - j: how many days back the return lies ($j = 0$ is day t); $\hat{\sigma}_t$: the estimate at the close of day t
  - **rolling window**: each day the newest return enters and the oldest leaves
- Zero-mean version: $\hat{\sigma}_t^2 = \frac{1}{n}\sum_{j=0}^{n-1} r_{t-j}^2$
  - for daily data the mean is tiny compared with σ (0.03% against 1.13% for the S&P 500); setting it to 0 removes a noisy estimate
- Usual windows: $n = 21$ (a month), 63 (a quarter), 252 (a year) of trading days
  - each $\hat{\sigma}_t$ is annualised with $\sqrt{A}$ and plotted at the last day of its window

S&P 500: Three Windows, Three Answers



- Rolling volatility of the S&P 500, annualised; dashed line: the whole-sample volatility, 18.0%
- **Question for the room**: which window would you trust on the day after a crash?

SFM_ch8_returns_volatility

Interpretation: the Window Trade-Off

- Short window ($n = 21$): reacts fast, but is noisy
  - maximum 97.5% on 6 April 2020; the typical daily change of $\ln \hat{\sigma}_t$ is 6.7%
- Long window ($n = 252$): smooth, but slow
  - maximum only 45.6%, reached on 15 July 2009, months after the peak of the 2008 crisis; daily change 0.8%
- **Answer**: a short window, or EWMA (next section), because a long window still averages the calm days before the crash
  - for capital planning, a long window is preferred: it changes slowly
- At the end of the sample: 9.6% (21 days), 11.1% (63 days), 12.9% (252 days)

SFM_ch8_returns_volatility

How Precise Is a Sample Volatility?

- With n i.i.d. Normal returns, the relative standard error of $\hat{\sigma}$ is $\mathrm{SE}(\hat{\sigma})/\sigma \approx 1/\sqrt{2n}$
  - $\mathrm{SE}(\hat{\sigma})$: the standard deviation of the estimate across samples; divided by σ it is a relative error
  - $n = 21$: 15.4%; $n = 63$: 8.9%; $n = 252$: 4.5%
- With heavy tails (kurtosis κ): $\mathrm{SE}(\hat{\sigma})/\sigma \approx \sqrt{(\kappa - 1)/(4n)}$ (delta method)
  - S&P 500: $\kappa = 13.9$, so $n = 21$: 39%; $n = 252$: 11%
  - for $\kappa = 3$ (Normal) the two formulas coincide
- **Worked example**: the last 21-day volatility of the S&P 500 is 9.6%
  - approximate 95% interval, Normal returns: [6.7%; 12.5%]; with the observed kurtosis: [2.2%; 17.0%]

1994: RiskMetrics Makes Volatility a Daily Number

- J.P. Morgan launched RiskMetrics in October 1994 and published its methods and data free of charge J.P. Morgan/Reuters (1996)
  - goal: one common yardstick for market risk across banks
  - volatilities and correlations of more than 480 series, updated every day
- The method: an exponentially weighted moving average (EWMA) of squared returns
  - one parameter, the **decay factor** λ: 0.94 for daily forecasts, 0.97 for monthly ones
  - for each series, λ minimises the root mean squared error of its variance forecasts; 0.94 is the weighted average over all series



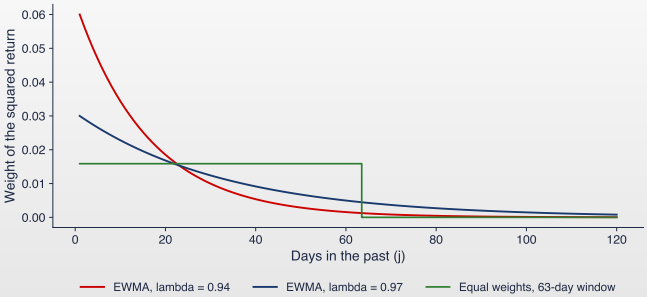
The J.P. Morgan building at 23 Wall Street, New York

Photo: Beyond My Ken (2012); CC BY-SA 4.0; Wikimedia Commons

The EWMA Estimator

- **EWMA** recursion (zero mean): $\sigma_t^2 = \lambda\,\sigma_{t-1}^2 + (1-\lambda)\,r_{t-1}^2,\ \ 0 < \lambda < 1$
  - today's variance = a weighted average of yesterday's variance and yesterday's squared return
  - λ: the weight of the past; a λ close to 1 gives a smooth, slowly reacting estimate
  - σ_t^2 uses returns up to day $t-1$: it is the forecast for day t
  - start: σ_1^2 = the mean of the first squared returns; its effect fades after a few months
- Unrolling the recursion: $\sigma_t^2 = (1-\lambda)\sum_{j\geq 1}\lambda^{j-1} r_{t-j}^2$
  - weights $(1-\lambda)\lambda^{j-1}$ decline geometrically and sum to 1
  - **half-life**: the lag at which the weight halves, $\ln 0.5 / \ln \lambda$
- A rolling window gives weight $1/n$ to each of the last n days and 0 to all older days

EWMA Weights against a Rolling Window



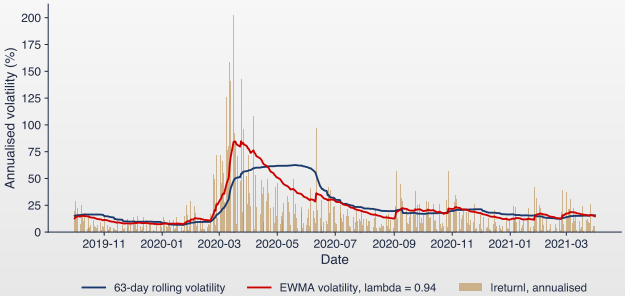
- $\lambda = 0.94$: half-life 11.2 days, mean lag $1/(1-\lambda) = 16.7$ days, 99% of the weight in the last 75 days
- $\lambda = 0.97$: half-life 22.8 days, 99% of the weight in the last 152 days

SFM_ch8_ewma

Worked Example: One EWMA Update

- Yesterday's EWMA variance: $\sigma^2_{t-1} = 1.00$ (%^{2}), a daily volatility of 1%; yesterday's return: $r_{t-1} = -3\%$
  - step 1: $\lambda\sigma^2_{t-1} = 0.94 \times 1.00 = 0.94$
  - step 2: $(1-\lambda)r^2_{t-1} = 0.06 \times 9 = 0.54$
  - step 3: $\sigma^2_t = 0.94 + 0.54 = 1.48$, so $\sigma_t = 1.217\%$ per day, $1.217 \times \sqrt{252} = 19.3\%$ per year
- If today is quiet ($r_t = 0$): $\sigma^2_{t+1} = 0.94 \times 1.48 = 1.39$
  - the shock fades by 6% per day; it never drops out abruptly
- A 63-day window would add $9/63 = 0.14$ to the variance today and remove this contribution at once, 63 days later

The Ghost Effect: March–June 2020



- S&P 500: 63-day rolling volatility and EWMA volatility ($\lambda = 0.94$), annualised; bars: $|r_t|$ on the same annual scale
- **Question for the room**: why does the blue line fall sharply in mid-June 2020, on a calm day?

SFM_ch8_ewma

Interpretation: Ghosts and Fading Memory

- **Answer**: on 15 June 2020 the return of 16 March 2020 (-12.8%) left the 63-day window
  - the rolling volatility fell from 50.4% to 43.0% in one day, with no news at all
  - this artefact is the **ghost effect**: an old shock leaves the window abruptly
- EWMA reacts faster and forgets smoothly
  - EWMA peak: 84.8% on 24 March 2020; 63-day peak: 62.6% only on 20 May 2020
  - on the ghost day EWMA changed by only -0.9 percentage points
- Price of the speed: EWMA is noisier than a long window and has no long-run level to return to

SFM_ch8_ewma

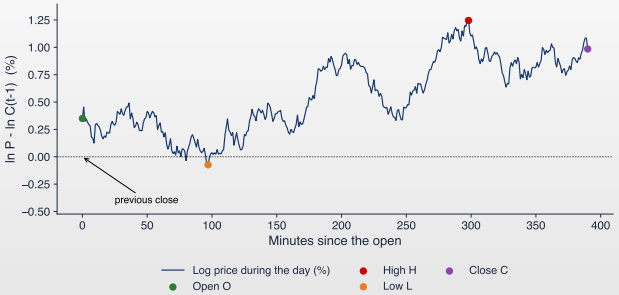
EWMA and GARCH: a Pointer to Chapter 9

- **ARCH** Engle (1982) and **GARCH** Bollerslev (1986): $\sigma_t^2 = \omega + \alpha\, r_{t-1}^2 + \beta\, \sigma_{t-1}^2$
  - $\omega > 0$: a constant; α: the reaction to yesterday's squared return; β: the weight of yesterday's variance
  - EWMA is the special case $\omega = 0$, $\alpha = 1 - \lambda$, $\beta = \lambda$, so $\alpha + \beta = 1$ ("integrated" GARCH, **IGARCH**)
- Consequences for forecasts made at day t for day $t + h$
  - EWMA: flat, $E_t[\sigma_{t+h}^2] = \sigma_{t+1}^2$ for every h (E_t: expectation given the information of day t): today's level persists for ever
  - GARCH with $\alpha + \beta < 1$: pulled towards the long-run variance $\omega/(1 - \alpha - \beta)$
- Chapter 9 estimates ω, α, β by maximum likelihood; here λ is fixed or chosen by a forecast criterion (Section 8)

Open, High, Low and Close Prices

- **OHLC** data: for each day t, the open O_t, the high H_t, the low L_t and the close C_t
  - the close-to-close return uses 2 prices a day; H_t and L_t carry information about the whole path of the day
- Notation (log differences in %), each from the open of day t:
  - $u_t = \ln(H_t/O_t)$, $d_t = \ln(L_t/O_t)$, $c_t = \ln(C_t/O_t)$; so $d_t \leq 0 \leq u_t$ and $d_t \leq c_t \leq u_t$
  - **overnight** return $o_t = \ln(O_t/C_{t-1})$; the daily return is $r_t = o_t + c_t$
  - the **range** $u_t - d_t = \ln(H_t/L_t)$
- Data: S&P 500 since 2008, DAX since 2006, Bitcoin since 2014, BVB stocks since 2010; the BET file has closes only

One Trading Day: What OHLC Records



- Simulated log price (Brownian motion, 390 one-minute steps) after an overnight gap of 0.35%: the close keeps only the end point; high and low summarise the path
- Two days with the same close can have very different ranges, and so very different risk

SFM_ch8_range_estimators

The Parkinson Estimator (1980)

- For a Brownian motion with no drift and variance σ^2 per day: $E[(u_t - d_t)^2] = 4 \ln 2 \, \sigma^2$ Parkinson (1980)
  - in words: the squared log range, rescaled so that on average it equals the variance
  - one day: $\hat{\sigma}^2_{P,t} = \dfrac{(u_t - d_t)^2}{4 \ln 2}$; on n days: the mean of these values
  - $4 \ln 2 \approx 2.77$: the range is on average larger than σ
- Assumptions
  - continuous trading during the day, so the observed high and low are the true extremes
  - no drift and no overnight jump: the range only sees the trading session
- Much more precise than r_t^2: a simulation later in this section measures by how much
  - the log range is also used to estimate stochastic volatility models Alizadeh, Brandt & Diebold (2002)

Garman–Klass (1980) and Rogers–Satchell (1991)

- **Garman–Klass** Garman & Klass (1980): adds the open-to-close return
  - $\hat{\sigma}^2_{GK,t} = 0.5\,(u_t - d_t)^2 - (2\ln 2 - 1)\,c_t^2$
  - the weights make the estimator unbiased and nearly of minimum variance among such quadratic forms, under no drift
- **Rogers–Satchell** Rogers & Satchell (1991): unbiased for any drift μ
  - $\hat{\sigma}^2_{RS,t} = u_t(u_t - c_t) + d_t(d_t - c_t)$
  - useful for assets with a strong trend over the window (for example, Bitcoin in a strong rally)
- Both still ignore the overnight return o_t: they measure the variance of the trading session only

Yang–Zhang (2000): Adding the Night

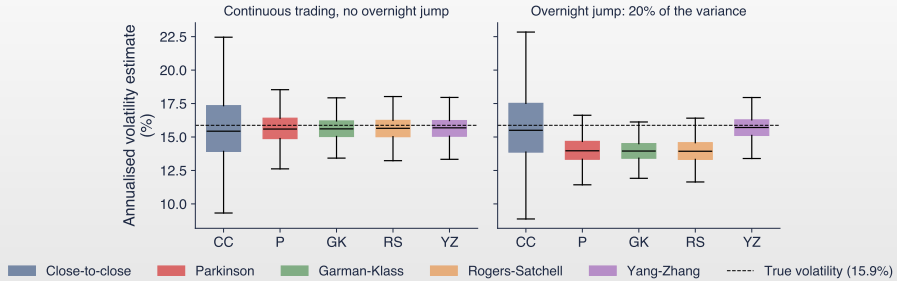
- On a window of n days Yang & Zhang (2000):
  - $\hat{\sigma}^2_{YZ} = \hat{\sigma}^2_O + k\,\hat{\sigma}^2_C + (1-k)\,\hat{\sigma}^2_{RS}, \quad k = \dfrac{0.34}{1.34 + (n+1)/(n-1)}$
  - $\hat{\sigma}^2_O$, $\hat{\sigma}^2_C$: sample variances of the overnight returns o_t and of the open-to-close returns c_t; $\hat{\sigma}^2_{RS}$: the mean Rogers–Satchell term
- Properties
  - unbiased with drift and with opening jumps; k minimises the variance of the estimator
  - $n = 21$: $k = 0.34/(1.34 + 22/20) = 0.34/2.44 = 0.14$
- Needs a window of $n \geq 2$ days (sample variances); Parkinson, Garman–Klass and Rogers–Satchell also work for a single day

Worked Example: S&P 500 on 16 March 2020

- Previous close 2 711; open 2 509, high 2 563, low 2 381, close 2 386
  - $o = -7.760$, $u = 2.145$, $d = -5.223$, $c = -5.005$ (in %); close-to-close $r = o + c = -12.765$
- One-day variance estimates ($\%^2$)
  - close-to-close: $r^2 = 163.0$
  - Parkinson: $(7.368)^2/2.773 = 19.6$; Garman–Klass: 17.5; Rogers–Satchell: 16.5
  - overnight part $o^2 = 60.2$; overnight plus Rogers–Satchell: 76.7
- Most of the fall happened overnight, before the open: estimators that ignore o_t miss most of the risk of this day

SFM_ch8_range_estimators

Efficiency by Simulation



- 1000 windows of 21 days; true volatility 1% per day (15.9% per year); intraday Brownian motion on 4680 steps (one every 5 seconds)
- **Efficiency** of an estimator: $\mathrm{Var}(\hat{\sigma}^2_{CC})/\mathrm{Var}(\hat{\sigma}^2)$, with CC the close-to-close estimator; right panel: 20% of the daily variance comes overnight

SFM_ch8_range_efficiency

Interpretation: Precision Is Not Accuracy

	CC	Parkinson	GK	RS	YZ
Efficiency, continuous trading	1.0	5.4	9.3	7.6	8.5
Bias (%), continuous trading	0	−2	−3	−3	−2
Bias (%), overnight jump	1	−22	−22	−22	−2
MSE ratio, overnight jump	1.0	1.8	1.8	1.8	9.0
Bias (%), 26 prices a day	0	−23	−32	−32	−28

- With continuous trading, range estimators are 5 to 9 times more efficient: one month of ranges carries as much information as 5 to 9 months of closes
- With an overnight jump, Parkinson, GK and RS miss the night (bias about −22%); only YZ stays unbiased; **MSE** (mean squared error) ratio = $\mathrm{MSE}(CC)/\mathrm{MSE}$
- With few trades the observed range is too small: all range estimators underestimate (thin markets) Molnár (2012)

SFM_ch8_range_efficiency

Real Data: Five Estimators, Five Assets

	Since	CC	P	GK	RS	YZ	Night share	$\mathrm{corr}(o, c)$
S&P 500	2008	19.9	15.4	14.6	14.4	16.3	13%	0.22
DAX	2006	20.9	16.7	16.6	16.6	20.3	31%	0.02
Bitcoin	2014	66.6	63.5	62.4	62.5	63.2	0%	0.02
TLV	2010	26.6	21.5	20.6	21.0	25.6	26%	−0.08
SNP	2010	24.5	21.4	20.8	21.4	25.8	27%	−0.20

- Annualised volatility (%) on the whole OHLC sample; P = Parkinson; night share $= \hat{\sigma}_O^2/(\hat{\sigma}_O^2 + \hat{\sigma}_C^2)$
- Stocks: Banca Transilvania (TLV) and OMV Petrom (SNP); open, high and low adjusted with the same factor as the close; days without trades removed

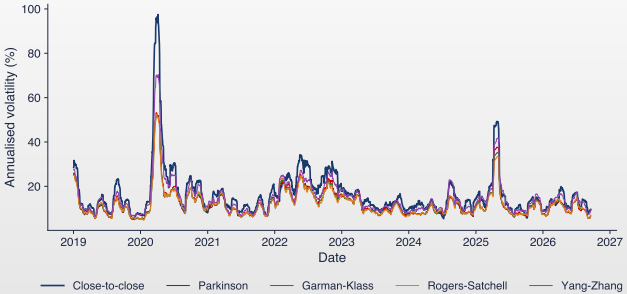
SFM_ch8_range_estimators

Interpretation: Why the Estimators Disagree

- Bitcoin trades 24 hours a day: the night share is almost 0, and all five estimators agree within a few points
  - the remaining gap: the daily path is observed in discrete trades, not continuously
- DAX and the BVB stocks: a quarter to a third of the variance comes overnight
  - Parkinson, GK and RS are far below CC; YZ, which adds the night, is close to CC
- S&P 500: even YZ (16.3%) stays below CC (19.9%)
  - overnight and open-to-close returns are positively correlated (0.22); a likely reason: the index open uses stale prices of stocks that have not traded yet
  - $\mathrm{Var}(r) = \mathrm{Var}(o) + \mathrm{Var}(c) + 2\,\mathrm{Cov}(o, c)$: no range estimator sees the covariance term
- Lesson: check the assumptions on the data before trusting an efficiency gain

SFM_ch8_range_estimators

S&P 500: Rolling 21-Day Estimates since 2019



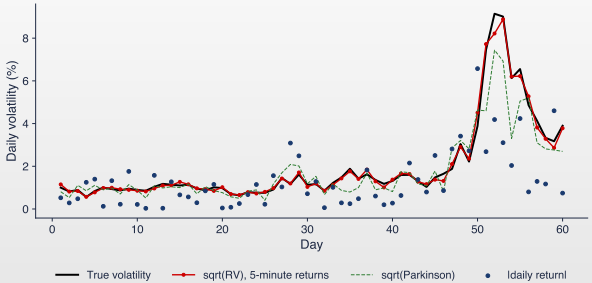
- Annualised 21-day volatility by the five estimators; means since 2019: CC 16.3%, Parkinson 12.5%, YZ 14.7%
- The range-based lines are smoother: the typical daily change of the log estimate is 4.4% for Parkinson and 4.4% for YZ, against 7.0% for CC

SFM_ch8_range_estimators

Realised Variance from Intraday Returns

- Split day t into M intervals (for example $M = 78$ five-minute returns $r_{t,i}$ in a 6.5-hour session)
  - $r_{t,i}$: the log return of interval i of day t, $i = 1, \ldots, M$
  - **realised variance** $\mathrm{RV}_t = \sum_{i=1}^{M} r_{t,i}^2$; **realised volatility** $= \sqrt{\mathrm{RV}_t}$
  - as $M \to \infty$, RV_t converges to the **integrated variance** of day t, the true variance of the day Andersen, Bollerslev, Diebold & Labys (2003), Barndorff-Nielsen & Shephard (2002)
- Volatility becomes almost observable: a nearly error-free measure for each day
  - used to evaluate forecasts (Section 8) and to forecast volatility directly, for example with the HAR model Corsi (2009)
  - high-frequency Bitcoin prices and daily risk: Pele & Mazurencu-Marinescu-Pele (2019)
- Our course data are daily; the next two slides use simulated one-minute prices to show how RV behaves

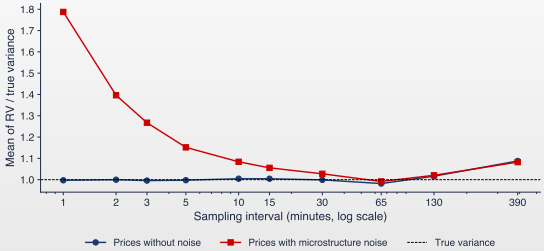
Simulated Example: RV Tracks the True Volatility



- 60 simulated days of 390 one-minute returns; the daily volatility follows a log AR(1) process (persistence 0.9)
- Mean absolute relative error: 5-minute RV 6%, Parkinson 23%, $|r_t|$ 58%; correlation with the truth: 0.995, 0.94 and 0.59

SFM_ch8_realised_volatility

Microstructure Noise and the Signature Plot



- Observed prices = true prices + **microstructure noise** (bid–ask bounce, price ticks), here with sd 0.025%; each squared return then adds 2× the noise variance
- With noise, one-minute RV overstates the variance by a factor of 1.79; at 5 minutes: 1.15; the **signature plot** shows RV against the sampling interval
- In practice, 5-minute RV is hard to beat Liu, Patton & Sheppard (2015)

SFM_ch8_realised_volatility

1982: Engle and the ARCH Model

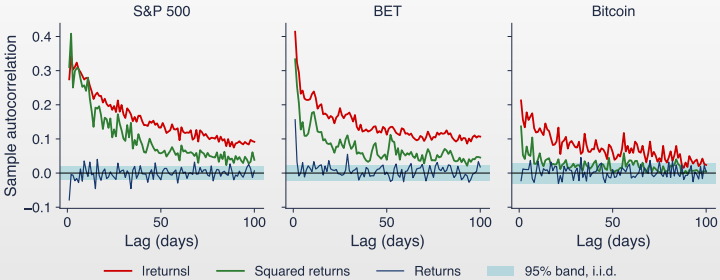
- **Volatility clustering**: periods of large absolute returns alternate with periods of small ones Mandelbrot (1963), Cont (2001)
  - returns are almost uncorrelated (Chapter 7), but $|r_t|$ and r_t^2 are strongly autocorrelated
- Robert F. Engle Engle (1982): the variance of today depends on yesterday's squared shocks: **ARCH**
  - the first model in which σ_t changes over time in a predictable way
  - Nobel Prize in Economic Sciences 2003, shared with Clive Granger Nobel Prize (2003); Engle's Nobel lecture: Engle (2004)
- This section: tests for clustering; Chapter 9: the models



Robert F. Engle (b. 1942), Santiago, 2017

Photo: Econterms (2017); CC BY-SA 4.0; Wikimedia Commons

The Sign Is Unpredictable, the Size Is Not



- Sample ACF (autocorrelation function, Chapter 7) of r_t, r_t^2 and $|r_t|$, lags 1–100, with the 95% band $\pm 1.96/\sqrt{T}$ for i.i.d. data
- S&P 500: all 100 autocorrelations of r_t^2 and of $|r_t|$ lie above the band; at lag 100, $|r_t|$ still has 0.09

SFM_ch8_clustering

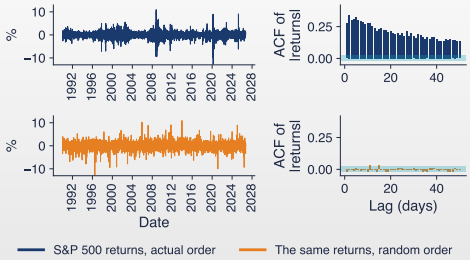
Interpretation: Clustering in Three Markets

	$\hat{\rho}_1(\lvert r\rvert)$	$\hat{\rho}_{10}(\lvert r\rvert)$	$\hat{\rho}_{100}(\lvert r\rvert)$	$\hat{\rho}_1(r^2)$	$\hat{\rho}_{10}(r^2)$	**Lags outside, $\lvert r\rvert$**
S&P 500	0.27	0.28	0.09	0.31	0.24	100/100
BET	0.41	0.23	0.11	0.33	0.18	100/100
Bitcoin	0.21	0.12	0.02	0.14	0.04	92/100

- Slow decay: a shock to volatility is still visible after months; Chapter 11 calls this long memory
- **Taylor effect**: $|r_t|$ is more autocorrelated than r_t^2 (96% of lags for the S&P 500, all lags for the BET and Bitcoin) Ding, Granger & Engle (1993)
- Bitcoin: weaker clustering in r_t^2 (26 of 100 lags outside the band): a few huge days dominate the squares

SFM_ch8_clustering

Is Clustering Real? Shuffle the Days



- The same S&P 500 returns in a random order: same distribution and same kurtosis (excess 10.9), but no clustering
- $\hat{\rho}_1(|r|)$: 0.27 in the actual order, -0.02 after shuffling; Ljung–Box $Q(10)$ of r_t^2: 8 015 against 6.3
- Clustering is a property of the **order** of the returns, not of their distribution

SFM_ch8_clustering

Testing for Clustering: Ljung–Box on Squared Returns

- H_0: no clustering, $\mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$ constant, so r_t^2 is uncorrelated
  - **McLeod–Li test** McLeod & Li (1983): the Ljung–Box statistic Ljung & Box (1978) applied to r_t^2
  - $Q(m) = T(T+2)\sum_{k=1}^{m} \hat{\rho}_k(r^2)^2/(T-k) \sim \chi^2(m)$ under H_0: a weighted sum of the first m squared autocorrelations
  - T: number of returns; m: number of lags tested; $\hat{\rho}_k(r^2)$: sample autocorrelation of r_t^2 at lag k
  - $Q(m) \geq 0$; large values mean autocorrelated squares; reject if $Q(m)$ exceeds the 95% quantile $\chi^2_{0.95}(m)$
- **Worked example**: S&P 500, $T = 9\,240$, $m = 3$
  - $\hat{\rho}_1, \hat{\rho}_2, \hat{\rho}_3$ of r_t^2: 0.311, 0.408, 0.250
  - terms $T(T+2)\hat{\rho}_k^2/(T-k)$: 892, 1 540, 579; $Q(3) = 3\,012 \gg \chi^2_{0.95}(3) = 7.81$: reject H_0
- The same test on r_t (Chapter 7) asks a different question: is the **mean** predictable?

SFM_ch8_clustering

The ARCH-LM Test (Engle, 1982)

- Auxiliary regression of the squared demeaned returns e_t^2, $e_t = r_t - \bar{r}$, on q of their own lags Engle (1982):
  - $e_t^2 = c_0 + c_1 e_{t-1}^2 + \cdots + c_q e_{t-q}^2 + u_t$; H_0: $c_1 = \cdots = c_q = 0$
  - $\bar{r}$: the sample mean; $c_0, \ldots, c_q$: regression coefficients; u_t: the error term
  - **LM** (Lagrange multiplier) statistic: $\mathrm{LM} = nR^2 \sim \chi^2(q)$, n = number of observations in the regression
- S&P 500, $q = 5$: $R^2 = 0.241$, $n = 9\,235$
  - $R^2 \in [0, 1]$: the share of the variance of e_t^2 explained by its own past
  - $\mathrm{LM} = 9\,235 \times 0.241 = 2\,227 \gg \chi^2_{0.95}(5) = 11.07$: strong ARCH effects
- Interpretation of R^2: about a quarter of the day-to-day variation of e_t^2 is predictable from the last five days
- Both tests are diagnostics: they say that clustering exists, not how to model it (Chapter 9)

SFM_ch8_clustering

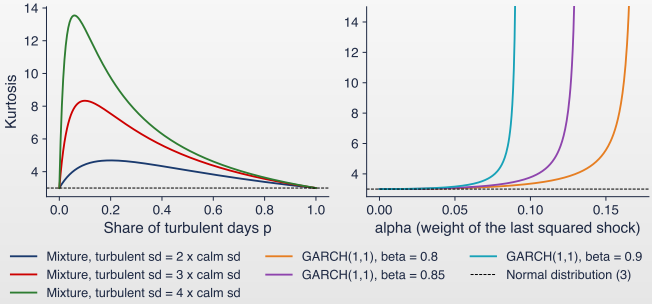
Clustering Tests in Six Series

	T	$\hat{\rho}_1(r)$	$Q(10)$, r	$\hat{\rho}_1(r^2)$	$Q(10)$, r^2	$Q(10)$, $\lvert r\rvert$	ARCH-LM(5)	Exc. kurt.
S&P 500	9 240	−0.079	90.8	0.311	8 015	8 323	2 227	10.9
DAX	9 288	−0.009	21.8	0.155	3 747	5 794	1 209	6.0
BET	7 243	0.156	198.3	0.333	2 500	5 184	1 057	9.9
Bitcoin	4 384	−0.022	20.2	0.137	213	1 086	115	12.0
TLV	3 788	0.010	29.1	0.122	481	1 179	253	13.5
SNP	3 815	−0.015	20.7	0.195	505	944	234	8.9

- 5% critical values: $\chi^2_{0.95}(10) = 18.31$ for $Q(10)$, $\chi^2_{0.95}(5) = 11.07$ for ARCH-LM(5); every statistic on r^2 and $|r|$ rejects with $p < 0.001$
- $Q(10)$ on r also rejects at 5% in all six series ($p \leq 0.028$), but it is far smaller than on r^2 (by a factor of 10.5 to 171.7): mean predictability is weak (Chapter 7), size predictability is strong everywhere
- Bitcoin, Banca Transilvania (TLV) and OMV Petrom (SNP) have the smallest ARCH-LM values despite heavy tails: $\mathrm{LM} = nR^2$ grows with the sample length, and these series are shorter; their R^2 is also lower (0.026, 0.067, 0.061 against 0.241 for the S&P 500): a few extreme days dilute the regression

SFM_ch8_clustering

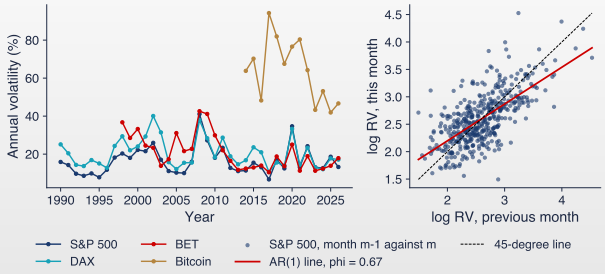
Clustering Creates Heavy Tails



- Left: calm days with sd 1, turbulent days with sd 2 (share $p = 0.2$): $E[\sigma^2] = 1.6$, $E[\sigma^4] = 4.0$, kurtosis $3E[\sigma^4]/E[\sigma^2]^2 = 4.69$, though each day is Normal
- Right: GARCH(1,1) kurtosis $3 + 6\alpha^2/(1 - \beta^2 - 2\alpha\beta - 3\alpha^2)$, as in SFEkurgarch; $\alpha = 0.10$, $\beta = 0.85$: 3.77 (Chapter 9)

SFM_ch8_kurgarch

Volatility over Three Decades



- Left: volatility of each calendar year (annualised, %); right: log monthly realised volatility of the S&P 500 (from daily returns) against the previous month
- S&P 500: from 7% (2017) to 41% (2008); BET: 10% (2017) to 43% (2008); Bitcoin: 42% (2025) to 94% (2017)

SFM_ch8_long_term_vix

Interpretation: Persistence and Mean Reversion

- Volatility changes over time, but it does not drift away: it returns to a normal level Schwert (1989)
  - AR(1) of log monthly realised volatility of the S&P 500 (441 months): $\hat{\phi} = 0.67$ (SE 0.04)
  - $\hat{\phi}$: the estimated weight of last month's log volatility; close to 1 means slow mean reversion
  - half-life of a volatility shock: $\ln 0.5 / \ln \hat{\phi} = 1.7$ months; long-run level about 13.5%
- Monthly volatility is right-skewed (skewness 3.3), its logarithm much less (0.6) Andersen, Bollerslev, Diebold & Ebens (2001)
  - mean 15.3% above the median 12.9%: a few crisis months pull the mean up
  - models of volatility are therefore often written for $\ln \sigma_t$
- Mean reversion is what EWMA lacks and GARCH with $\alpha + \beta < 1$ has (Chapter 9)

SFM_ch8_long_term_vix

The VIX: Implied Volatility

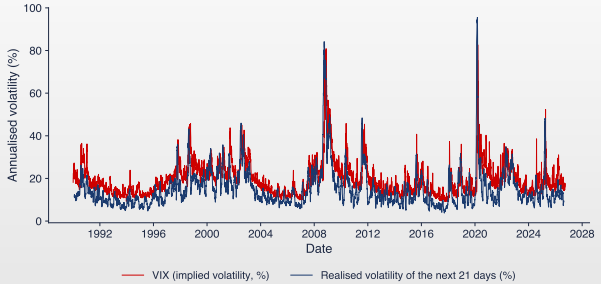
- Options trade on exchanges since 1973, when members of the Chicago Board of Trade founded the **CBOE** (Chicago Board Options Exchange)
  - an option price depends on the volatility the market expects until expiry
  - **implied volatility**: the volatility that, put into an option-pricing formula, gives the observed price
- **VIX**, introduced in 1993 Whaley (1993): the expected volatility of the S&P 500 over the next 30 calendar days, from S&P 500 option prices Cboe (2026)
  - quoted in annualised %: VIX $= 20$ means about $20/\sqrt{12} \approx 5.8\%$ over a month
  - called the "fear index": it jumps when stocks fall Whaley (2009)



Traders at the Chicago Board of Trade, 31 May 1973

Photo: U.S. Department of Agriculture (1973); public domain; Wikimedia Commons

VIX against the Volatility That Followed



- VIX and the realised volatility of the S&P 500 over the next 21 trading days, $\sqrt{(A/21)\sum_{j=1}^{21} r_{t+j}^2}$; maximum VIX: 82.7 on 16 March 2020
- **Question for the room**: is the red line usually above or below the blue one?

SFM_ch8_long_term_vix

Interpretation: Implied against Realised Volatility

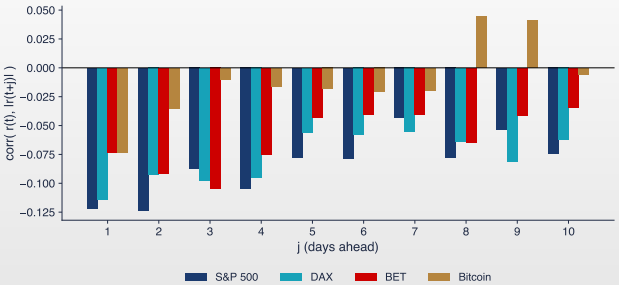
- **Answer**: above; VIX exceeded the next 21 days' realised volatility on 86% of the 9 199 days
  - mean VIX 19.4% against mean realised 15.3%: a gap of 4.1 points
  - **variance risk premium**: investors pay to be insured against volatility spikes Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009)
- As a forecast, VIX is informative but biased
  - regression of future on implied: realised $= -1.72 + 0.88\,\text{VIX}$, $R^2 = 0.52$
  - the past 21-day realised volatility explains less: $R^2 = 0.44$
- Daily log changes of the VIX and S&P 500 returns: correlation -0.71: fear rises when prices fall

SFM_ch8_long_term_vix

The Leverage Effect

- **Leverage effect**: negative returns are followed by higher volatility than positive returns of the same size Christie (1982)
  - leverage story: a falling share price raises the debt-to-equity ratio, so the equity becomes riskier
  - volatility-feedback story: higher expected volatility raises the required return, so the price falls today French, Schwert & Stambaugh (1987), Bekaert & Wu (2000)
- A simple measure: $\mathrm{corr}(r_t, |r_{t+j}|)$ for $j = 1, 2, \ldots$
  - j: the number of days ahead; corr: the sample correlation over all days t
  - negative: today's fall announces tomorrow's turbulence; zero under a symmetric model
- Chapter 2 met the asymmetry of the distribution; here the asymmetry is in the **dynamics** of volatility

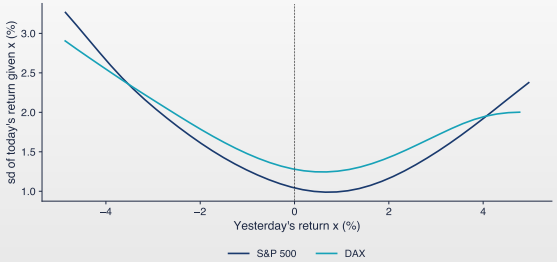
Bad News Raises Future Volatility



- $\mathrm{corr}(r_t, |r_{t+j}|)$, $j = 1, \ldots, 10$; S&P 500: -0.122 at $j = 1$; the reverse direction $\mathrm{corr}(|r_{t-1}|, r_t) = 0.025$
- Mean r^2_{t+1} after a down day divided by that after an up day: S&P 500 1.51, DAX 1.46, BET 1.41, Bitcoin 1.06

SFM_ch8_leverage

Nonparametric Volatility Given Yesterday's Return



- $\hat{\sigma}(x) = \sqrt{\hat{m}_{\mathbf{2}}(x) - \hat{m}_{\mathbf{1}}(x)^{\mathbf{2}}}$, $\hat{m}_{\mathbf{1}}$, $\hat{m}_{\mathbf{2}}$: local linear regressions of r_t and $r_t^{\mathbf{2}}$ on $r_{t-\mathbf{1}} = x$ (quartic kernel, bandwidth 0.04 on decimal returns, as in SFEvolnonparest)
- S&P 500: $\hat{\sigma}(-2\%) = 1.61\%$ against $\hat{\sigma}(+2\%) = 1.16\%$; the minimum is at $x \approx 0.7\%$, not at 0: an asymmetric news impact

SFM_ch8_leverage

The Proxy Problem

- A forecast h_t of σ_t^2 is made at the close of day $t-1$; the true σ_t^2 is never observed
  - we compare h_t with a **proxy** $\hat{\sigma}_t^2$: a measurable quantity with $E[\hat{\sigma}_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2$
- Proxies, from noisy to precise: r_t^2, a range estimator, realised variance
  - with r_t^2 even a perfect forecast looks poor: Andersen & Bollerslev (1998) showed that a low R^2 does not mean a bad model
- **Mincer–Zarnowitz regression** Mincer & Zarnowitz (1969): $\hat{\sigma}_t^2 = a + b\, h_t + u_t$
  - a: intercept; b: slope; u_t: error; a good forecast has $a \approx 0$, $b \approx 1$
  - the R^2 measures how much of the proxy the forecast explains

Loss Functions: MSE and QLIKE

- Average loss over the evaluation days; lower is better Patton (2011)
  - L: the loss of one day; h_t: the variance forecast; $\hat{\sigma}_t^2$: the proxy
  - MSE: $L = (\hat{\sigma}_t^2 - h_t)^2$
  - **QLIKE** (quasi-likelihood): $L = \hat{\sigma}_t^2 / h_t + \ln h_t$, minus the Normal log-likelihood up to constants
- Both are **robust**: with a noisy but unbiased proxy they rank forecasts as the true variance would
  - losses on $\sqrt{h_t}$ or on $\ln h_t$ are not robust and can prefer the wrong forecast
- **Worked example**: true variance and proxy equal to 1; two forecasts, $h = 0.5$ and $h = 2$
  - MSE: $(1 - 0.5)^2 = 0.25$ against $(1 - 2)^2 = 1$: MSE punishes over-prediction more
  - QLIKE: $1/0.5 + (-0.693) = 1.307$ against $1/2 + 0.693 = 1.193$: QLIKE punishes under-prediction more, the costly error for risk management

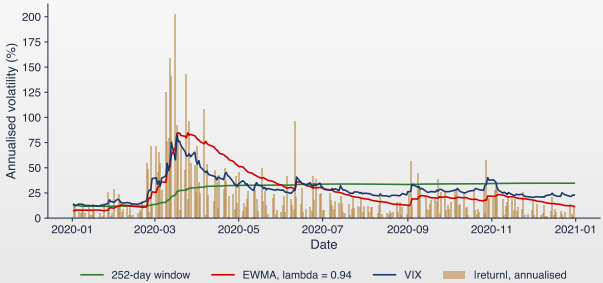
S&P 500: Which Forecast Is Best?

	21 days	63 days	252 days	EWMA 0.94	EWMA 0.97	VIX
MSE	18.99	20.77	21.43	17.77	19.02	16.70
QLIKE	0.876	0.914	1.099	0.808	0.844	0.799
MZ R^2, proxy r^2	0.14	0.05	0.01	0.17	0.11	0.25
MZ R^2, range proxy	0.24	0.09	0.02	0.29	0.19	0.42

- Next-day variance forecasts for 4 202 days from 2010; proxy r_t^2; VIX forecast: VIX^2/A; range proxy: o_t^2 plus the Garman–Klass term
- **DM** (Diebold–Mariano) test of equal QLIKE Diebold & Mariano (1995): EWMA 0.94 against 21 days $t = -4.50$ ($p < 0.001$); VIX against EWMA 0.94 $t = -0.32$ ($p = 0.751$)
- EWMA beats the 21-day window; VIX and EWMA are not significantly different; the long window is worst; a precise proxy raises every R^2, but not the ranking

SFM_ch8_forecast_evaluation

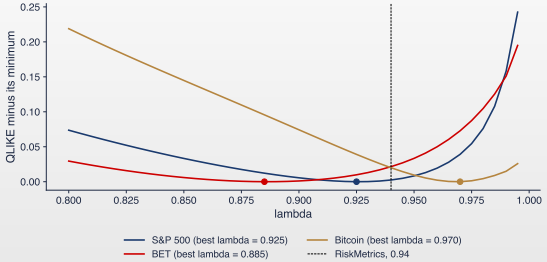
Forecasts in 2020



- Next-day volatility forecasts of the S&P 500 in 2020 (annualised): 252-day window, EWMA ($\lambda = 0.94$) and VIX; bars: $|r_t|$
- VIX started to rise one trading day before the first large fall of February 2020; EWMA rose right after it; the 252-day window moved slowly and stayed near its peak for the rest of the year

SFM_ch8_forecast_evaluation

Choosing λ by QLIKE



- Mean QLIKE of next-day EWMA forecasts (proxy r_t^2, from 2010; Bitcoin from 2016) minus its minimum, for λ from 0.80 to 0.995
- Best λ: S&P 500 0.925, BET 0.885, Bitcoin 0.970; the RiskMetrics 0.94 loses only 0.002 for the S&P 500
- One λ for all assets is a convenient compromise, not an optimum

SFM_ch8_forecast_evaluation

An Open Question

- **Do range-based estimators help on the Bucharest Stock Exchange, where many stocks trade thinly?**
  - theory: ranges are 5 to 9 times more efficient; but thin trading shrinks the observed range (Section 4)
  - days with trades but an unchanged close, since 2010: TLV 9%, SNP 8%, BRD 10%, TGN 10%
  - Parkinson / close-to-close volatility: TLV 0.81, SNP 0.87, BRD 0.90, TGN 0.87 (S&P 500: 0.78)
  - QLIKE of next-day forecasts by the 21-day window, CC against YZ: TLV 2.022 against 2.039; TGN 1.750 against 1.636
- Open: the answer differs by stock; is it liquidity, the opening auction, or the size of the overnight move?
- AI tools can speed up such a study, but its results must still be checked Wang et al. (2023)

SFM_ch8_bvb_range

How AI Could Help

- **Literature**: list studies of range-based and realised volatility in emerging and thin markets, with their data and estimators
- **Code**: a first draft of the five estimators, rolling forecasts and QLIKE comparisons for all BET stocks
- **Robustness**: liquidity groups by volume, other windows, the 2016–2020 and 2021–2026 halves
- Example prompt
  - `Write a Python function that computes the Parkinson, Garman-Klass, Rogers-Satchell and Yang-Zhang variance on rolling 21-day windows from daily open, high, low and close prices, and compares their next-day forecasts with QLIKE against squared close-to-close returns.`

What to Check

- The constants: $4 \ln 2$ in Parkinson, $2 \ln 2 - 1$ in Garman–Klass, k in Yang–Zhang; a draft with $\ln 2$ or $k = 0.34$ is wrong
- Stock prices: open, high and low must be adjusted for dividends and splits with the same factor as the close
- Annualisation with the actual frequency: 365 for crypto, about 250 for BVB stocks
- No look-ahead: the forecast for day t uses only data up to day $t - 1$
- References: every cited paper must exist; check the DOI

Project Idea

- **Question**: which daily volatility estimator should a Romanian risk manager use?
  - data: BVB stocks since 2010 (TLV, SNP, BRD, TGN, and others with OHLC data), S&P 500 and DAX as benchmarks (EODHD)
- Steps
  - the five estimators on 21-day windows; EWMA versions of each; next-day forecasts
  - QLIKE and MSE with the proxy r_t^2; Diebold–Mariano tests; results by liquidity group
  - relate the gain of YZ over CC to the share of zero-volume days and to the overnight share
- Deliverables: one table, one chart, and a paragraph on what the data can and cannot show
- Declare any AI use, and list the errors of the AI that you corrected

Key Takeaways

- Volatility is the standard deviation of returns; it is latent and changes over time; annualise with the actual frequency
- Historical windows trade speed against noise; EWMA ($\lambda = 0.94$) reacts fast and avoids ghosts
- Range estimators use the daily path: far more efficient, but biased by overnight jumps and thin trading; Yang–Zhang adds the night
- Clustering: $|r_t|$ and r_t^2 are autocorrelated; detect it with Ljung–Box on r_t^2 and ARCH-LM
- Volatility mean-reverts; VIX usually exceeds realised volatility; falls raise future volatility
- Compare forecasts with robust losses (QLIKE) and a proxy; test the differences

Key Formulas

Quantity	**Formula**
Annualised volatility	$\sigma_{\text{year}} = \sigma_{\text{day}}\sqrt{A}$
EWMA	$\sigma_t^2 = \lambda\sigma_{t-1}^2 + (1-\lambda)r_{t-1}^2$, half-life $\ln 0.5/\ln\lambda$
Parkinson	$(u_t - d_t)^2/(4\ln 2)$
Garman–Klass	$0.5(u_t - d_t)^2 - (2\ln 2 - 1)c_t^2$
Rogers–Satchell	$u_t(u_t - c_t) + d_t(d_t - c_t)$
Yang–Zhang	$\hat\sigma_O^2 + k\hat\sigma_C^2 + (1-k)\hat\sigma_{RS}^2$, $k = 0.34/(1.34 + (n+1)/(n-1))$
Realised variance	$\mathrm{RV}_t = \sum_i r_{t,i}^2$
McLeod–Li	$Q(m) = T(T+2)\sum_{k=1}^{m}\hat\rho_k(r^2)^2/(T-k) \sim \chi^2(m)$
ARCH-LM	$\mathrm{LM} = nR^2 \sim \chi^2(q)$
MSE, QLIKE	$(\hat\sigma_t^2 - h_t)^2$, $\hat\sigma_t^2/h_t + \ln h_t$

Check Yourself

- **Question**: a stock has $H = 102$, $L = 98$ on one day; what is the Parkinson daily volatility?
  - **Answer**: $\ln(102/98) = 4.00\%$; $\sqrt{4.00^2/2.77} = 2.40\%$
- **Question**: Ljung–Box on returns does not reject, on squared returns it rejects strongly; what do you conclude?
  - **Answer**: the mean is unpredictable, the variance is predictable: volatility clustering without autocorrelation
- **Question**: why can EWMA with $\lambda = 0.94$ not forecast a return of volatility to its normal level?
  - **Answer**: $\alpha + \beta = 1$ and $\omega = 0$: its forecasts are flat at every horizon
- Next: Chapter 9, ARCH and GARCH models

References (1/3)

- Alizadeh, S., Brandt, M.W., Diebold, F.X. (2002). Range-based estimation of stochastic volatility models. *Journal of Finance*, 57(3), 1047–1091.
- Andersen, T.G., Bollerslev, T. (1998). Answering the skeptics: yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 39(4), 885–905.
- Andersen, T.G., Bollerslev, T., Diebold, F.X., Ebens, H. (2001). The distribution of realized stock return volatility. *Journal of Financial Economics*, 61(1), 43–76.
- Andersen, T.G., Bollerslev, T., Diebold, F.X., Labys, P. (2003). Modeling and forecasting realized volatility. *Econometrica*, 71(2), 579–625.
- Barndorff-Nielsen, O.E., Shephard, N. (2002). Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society, Series B*, 64(2), 253–280.
- Bekaert, G., Wu, G. (2000). Asymmetric volatility and risk in equity markets. *Review of Financial Studies*, 13(1), 1–42.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- Bollerslev, T., Tauchen, G., Zhou, H. (2009). Expected stock returns and variance risk premia. *Review of Financial Studies*, 22(11), 4463–4492.
- Borak, S., Härdle, W.K., López-Cabrera, B. (2013). *Statistics of Financial Markets: Exercises and Solutions*, 2nd ed. Springer.
- Carr, P., Wu, L. (2009). Variance risk premiums. *Review of Financial Studies*, 22(3), 1311–1341.
- Cboe Global Markets (2026). Cboe Volatility Index (VIX). Product page and methodology.
- Christie, A.A. (1982). The stochastic behavior of common stock variances: value, leverage and interest rate effects. *Journal of Financial Economics*, 10(4), 407–432.
- Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196.

References (2/3)

- Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- Ding, Z., Granger, C.W.J., Engle, R.F. (1993). A long memory property of stock market returns and a new model. *Journal of Empirical Finance*, 1(1), 83–106.
- Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- Engle, R.F. (2004). Risk and volatility: econometric models and financial practice. *American Economic Review*, 94(3), 405–420.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- French, K.R., Schwert, G.W., Stambaugh, R.F. (1987). Expected stock returns and volatility. *Journal of Financial Economics*, 19(1), 3–29.
- Garman, M.B., Klass, M.J. (1980). On the estimation of security price volatilities from historical data. *Journal of Business*, 53(1), 67–78.
- Hansen, P.R., Lunde, A. (2005). A forecast comparison of volatility models: does anything beat a GARCH(1,1)? *Journal of Applied Econometrics*, 20(7), 873–889.
- J.P. Morgan/Reuters (1996). *RiskMetrics – Technical Document*, 4th ed. New York: Morgan Guaranty Trust Company.
- Liu, L.Y., Patton, A.J., Sheppard, K. (2015). Does anything beat 5-minute RV? A comparison of realized measures across multiple asset classes. *Journal of Econometrics*, 187(1), 293–311.
- Ljung, G.M., Box, G.E.P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2), 297–303.
- Mandelbrot, B. (1963). The variation of certain speculative prices. *Journal of Business*, 36(4), 394–419.
- McLeod, A.I., Li, W.K. (1983). Diagnostic checking ARMA time series models using squared-residual autocorrelations. *Journal of Time Series Analysis*, 4(4), 269–273.
- Mincer, J.A., Zarnowitz, V. (1969). The evaluation of economic forecasts. In J.A. Mincer (ed.), *Economic Forecasts and Expectations*, 3–46. NBER.

References (3/3)

- Molnár, P. (2012). Properties of range-based volatility estimators. *International Review of Financial Analysis*, 23, 20–29.
- Nobel Prize Outreach (2003). The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2003: Robert F. Engle III and Clive W.J. Granger. nobelprize.org.
- Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *Journal of Business*, 53(1), 61–65.
- Patton, A.J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256.
- Pele, D.T., Lazar, E., Dufour, A. (2017). Information entropy and measures of market risk. *Entropy*, 19(5), 226.
- Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019). Using high-frequency entropy to forecast Bitcoin's daily Value at Risk. *Entropy*, 21(2), 102.
- Poon, S.-H., Granger, C.W.J. (2003). Forecasting volatility in financial markets: a review. *Journal of Economic Literature*, 41(2), 478–539.
- Rogers, L.C.G., Satchell, S.E. (1991). Estimating variance from high, low and closing prices. *Annals of Applied Probability*, 1(4), 504–512.
- Schwert, G.W. (1989). Why does stock market volatility change over time? *Journal of Finance*, 44(5), 1115–1153.
- Tsay, R.S. (2010). *Analysis of Financial Time Series*, 3rd ed. Wiley.
- Wang, H., Fu, T., Du, Y., Gao, W., et al. (2023). Scientific discovery in the age of artificial intelligence. *Nature*, 620, 47–60.
- Whaley, R.E. (1993). Derivatives on market volatility. *Journal of Derivatives*, 1(1), 71–84.
- Whaley, R.E. (2009). Understanding the VIX. *Journal of Portfolio Management*, 35(3), 98–105.
- Yang, D., Zhang, Q. (2000). Drift-independent volatility estimation based on high, low, open, and close prices. *Journal of Business*, 73(3), 477–492.